

EX. NO: 3

IMPLEMENTATION OF POLYALPHABETIC CIPHER

AIM:

To write a C program to implement the Polyalphabetic Substitution Cipher technique.

ALGORITHM:

1. Start the program.
2. Enter the plain text.
3. Enter the key.
4. Repeat the key to match the length of the plain text.
5. Read each character of the plain text and the corresponding key character.
6. Add the values of the plain text character and key character.
7. Take the result modulo 26 to obtain the encrypted character.
8. Keep spaces and special characters unchanged.
9. Display the encrypted text.
10. Stop the program.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
void main()
```

```
{
```

```
    char plain[100], key[100], cipher[100];
```

```
    int i, j = 0, keylen;
```

```
printf("Enter the plain text: ");
```

```
fgets(plain, sizeof(plain), stdin);
```

```
printf("Enter the key: ");
```

```
scanf("%s", key);
```

```
keylen = strlen(key);
```

```
for(i = 0; plain[i] != '\0'; i++)
```

```
{
```

```
    if(isalpha(plain[i]))
```

```
    {
```

```
        if(isupper(plain[i]))
```

```
            cipher[i] = ((plain[i] - 'A') +
```

```
                        (toupper(key[j % keylen]) - 'A')) % 26 + 'A';
```

```
        else
```

```
            cipher[i] = ((plain[i] - 'a') +
```

```
                        (tolower(key[j % keylen]) - 'a')) % 26 + 'a';
```

```

        j++;

    }

    else

    {

        cipher[i] = plain[i];

    }

}

cipher[i] = '\0';

printf("\nPLAIN TEXT: %s", plain);

printf("ENCRYPTED TEXT: %s", cipher);

}

```

OUTPUT:

The screenshot shows an online compiler interface with a dark theme. The code editor on the left contains C code for a Caesar cipher. The output panel on the right shows the program's execution results.

Code Editor:

```

25         else
26             cipher[i] = ((plain[i] - 'a' +
27                           (tolower(key[j % keylen]) - 'a')) % 26 +
28                           'a');
29         j++;
30     }
31     else
32     {
33         cipher[i] = plain[i];
34     }
35 }
36
37 cipher[i] = '\0';
38
39 printf("\nPLAIN TEXT: %s", plain);

```

Output Panel:

```

Enter the plain text: APPLE IS RED
Enter the key: GOD

PLAIN TEXT: APPLE IS RED
ENCRYPTED TEXT: GDSRS LY FHJ

=== Code Exited With Errors ===

```

RESULT:

Thus, the Polyalphabetic Substitution Cipher was successfully implemented using C language, and the given plaintext was encrypted using the given key.